

All-aperture self-adjointness for the MIT bag model on an unbounded circular cone

Candidate full solution packet for arXiv:2201.08192

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Abstract

Cassano and Lotoreichik proved essential self-adjointness of the MIT bag Dirac operator on convex circular cones and conjectured it for every aperture. Their reduction leaves one Ferrers-function equation for the re-entrant range. We prove that the left side minus the right side of this equation is strictly positive throughout the critical spectral interval. The Mehler–Dirichlet formula turns the expression into an integral which, for nonnegative spectral parameter, is minimized at the endpoint $\lambda = 1/2$. That endpoint reduces to a strict elementary inequality between complete elliptic integrals. This closes the last spectral case and proves the conjecture for all $0 < \omega < \pi$.

1 Source conjecture

The source is B. Cassano and V. Lotoreichik, “Self-adjointness for the MIT bag model on an unbounded cone,” arXiv:2201.08192, later published in *Mathematische Nachrichten* 297 (2024), 1006–1041 [1]. On physical PDF page 6, Remark 2.4 states:

“We conjecture that Theorem 2.3 is true for all the values $\omega \in (0, \pi)$ of the aperture angle. In Appendix C we provide numerical evidence to support our conjecture, showing that for $\pi/2 < \omega < \pi$ all the eigenvalues of the spin-orbit operator K_ω escape the critical interval $[-1/2, 1/2]$. This implies essential self-adjointness of D_ω and the validity of (2.9) for all $\omega \in (0, \pi)$.”

The rendered source-page evidence, including the reduction immediately before the conjecture, is reproduced below.

The proof of Theorem 2.3 is based on the decomposition in Theorem 2.1 and reduces to showing that the eigenvalues of the spin-orbit operator K_ω escape the critical interval $[-\frac{1}{2}, \frac{1}{2}]$ or, equivalently, that $\sigma(T_k) \cap [-\frac{1}{2}, \frac{1}{2}] = \emptyset$ for all $k \in \mathbb{Z}$.

Remark 2.4. We conjecture that Theorem 2.3 is true for all the values $\omega \in (0, \pi)$ of the aperture angle. In Appendix C we provide numerical evidence to support our conjecture, showing that for $\pi/2 < \omega < \pi$ all the eigenvalues of the spin-orbit operator K_ω escape the critical interval $[-\frac{1}{2}, \frac{1}{2}]$. This implies essential self-adjointness of D_ω and the validity of (2.9) for all $\omega \in (0, \pi)$.

Remark 2.5. The operator K_ω can be viewed as a Dirac-type operator on the manifold $\mathbb{S}^2 \cap \mathcal{C}_\omega$. Geometric spectral bounds in the spirit of [HMZ01] can also be useful to show that there are no eigenvalues of K_ω in the interval $[-\frac{1}{2}, \frac{1}{2}]$. This approach is also suitable to deal with general non-circular cones, for which separation of variables can not be used any more.

Remark 2.6. In the literature, sometimes also the following operator D_ω^- is called Dirac operator with MIT bag boundary conditions:

$$(2.10) \quad \begin{aligned} D_\omega^- u &:= \mathcal{D}u, \\ \text{dom } D_\omega^- &:= \{u \in C_0^\infty(\overline{\mathcal{C}_\omega} \setminus \mathbf{0}; \mathbb{C}^4) : u|_{\partial \mathcal{C}_\omega} - i\beta(\alpha \cdot \nu_\omega)u|_{\partial \mathcal{C}_\omega} = 0\}, \end{aligned}$$

Here D_ω is the massless Dirac operator on the circular cone $\mathcal{C}_\omega$ with MIT bag boundary conditions, initially defined away from the vertex. The conjectured domain of its closure is

$$\text{dom } \overline{D_\omega} = \{u \in H^1(\mathcal{C}_\omega; \mathbb{C}^4) : u|_{\partial \mathcal{C}_\omega} + i\beta(\alpha \cdot \nu_\omega)u|_{\partial \mathcal{C}_\omega} = 0\}. \quad (1)$$

2 The exact missing scalar statement

Write $P_\nu^\mu(x)$ for the Ferrers function of the first kind and $P_\nu = P_\nu^0$. The source proves the conjecture for $0 < \omega < \pi/2$ and records that the half-space case $\omega = \pi/2$ is known. For $\pi/2 < \omega < \pi$, Appendix C makes the remaining reduction explicit:

- all angular modes $k \notin \{-1, 0\}$ already have $\sigma(T_k) \cap [-1/2, 1/2] = \emptyset$;
- the $k = -1$ and $k = 0$ cases are equivalent;
- the relevant $k = 0$ set consists of the roots of

$$(\lambda + 1)P_\lambda^{-1}(\cos \omega) = P_{\lambda-1}(\cos \omega). \quad (2)$$

Thus it suffices to prove that (2) has no root in $[-1/2, 1/2]$ for a re-entrant aperture.

3 Proof intuition

The source equation looks delicate because the two Ferrers functions can each change sign near $x = -1$. A recurrence combines them before any sign estimate is made. The Mehler–Dirichlet formula then writes the combined expression as an integral whose dependence on λ is elementary trigonometry. Negative λ gives a pointwise positive integrand. For $0 \leq \lambda \leq 1/2$, that integrand decreases with λ , so the entire interval reduces to $\lambda = 1/2$. A half-angle substitution evaluates the endpoint using complete elliptic integrals E and K . Pairing the integrands at y and $\pi/2 - y$ proves the needed strict inequality $E > qK$.

4 Ferrers positivity lemma

Lemma 1. *For every $\theta \in (\pi/2, \pi)$ and every $\lambda \in [-1/2, 1/2]$,*

$$F_\theta(\lambda) := (\lambda + 1)P_\lambda^{-1}(\cos \theta) - P_{\lambda-1}(\cos \theta) > 0. \quad (3)$$

Proof. Put

$$x = \cos \theta < 0, \quad s = \sin \theta > 0, \quad p = \sin \frac{\theta}{2}, \quad q = \cos \frac{\theta}{2}.$$

Then $p > q > 0$. The degree symmetry $P_{\lambda-1} = P_{-\lambda}$ and a standard Ferrers recurrence give

$$(\lambda + 1)P_\lambda^{-1}(x) = \frac{P_{-\lambda}(x) - xP_\lambda(x)}{s}. \quad (4)$$

The identity is regular also at $\lambda = 0$. Consequently,

$$sF_\theta(\lambda) = (1 - s)P_{-\lambda}(x) - xP_\lambda(x) =: H_\theta(\lambda). \quad (5)$$

For $0 < \theta < \pi$, the Mehler–Dirichlet formula is

$$P_\nu(\cos \theta) = \frac{\sqrt{2}}{\pi} \int_0^\theta \frac{\cos((\nu + \frac{1}{2})t)}{\sqrt{\cos t - \cos \theta}} dt \quad [2, \text{Eq. 14.12.1}]. \quad (6)$$

Using

$$1 - s - x = 2p(p - q), \quad 1 - s + x = -2q(p - q),$$

and expanding the two cosines in (6), equation (5) becomes

$$H_\theta(\lambda) = \frac{2\sqrt{2}(p - q)}{\pi} \int_0^\theta \frac{J_\lambda(t)}{\sqrt{\cos t - \cos \theta}} dt, \quad (7)$$

where

$$J_\lambda(t) = p \cos \frac{t}{2} \cos(\lambda t) - q \sin \frac{t}{2} \sin(\lambda t). \quad (8)$$

If $-1/2 \leq \lambda \leq 0$, then $\cos(\lambda t) > 0$ and $\sin(\lambda t) \leq 0$ for $0 < t < \theta < \pi$. Hence $J_\lambda(t) > 0$ pointwise, and (7) is positive.

Suppose now that $0 \leq \lambda \leq 1/2$. For $0 < t < \theta$,

$$\frac{\partial J_\lambda(t)}{\partial \lambda} = -t \left[p \cos \frac{t}{2} \sin(\lambda t) + q \sin \frac{t}{2} \cos(\lambda t) \right] \leq 0. \quad (9)$$

Therefore $J_\lambda(t) \geq J_{1/2}(t)$ and it remains only to prove that the integral of

$$J_{1/2}(t) = p \cos^2 \frac{t}{2} - q \sin^2 \frac{t}{2}$$

is positive.

Let $u = \sin(t/2)/p$. With the standard complete elliptic integrals

$$K(p) = \int_0^{\pi/2} \frac{dy}{\sqrt{1 - p^2 \sin^2 y}}, \quad E(p) = \int_0^{\pi/2} \sqrt{1 - p^2 \sin^2 y} dy,$$

the elementary substitution, followed by $u = \sin y$, gives

$$\begin{aligned} \int_0^\theta \frac{J_{1/2}(t)}{\sqrt{\cos t - \cos \theta}} dt &= \sqrt{2} \int_0^1 \frac{p - p^2(p+q)u^2}{\sqrt{1-u^2}\sqrt{1-p^2u^2}} du \\ &= \sqrt{2}[(p+q)E(p) - qK(p)]. \end{aligned} \quad (10)$$

We finish with a short proof that $E(p) > qK(p)$. Set

$$a(y) = \sqrt{1 - p^2 \sin^2 y}, \quad b(y) = a(\frac{\pi}{2} - y) = \sqrt{1 - p^2 \cos^2 y}.$$

Because $p^2 + q^2 = 1$,

$$a(y)^2 b(y)^2 = q^2 + p^4 \sin^2 y \cos^2 y \geq q^2.$$

Pairing y with $\pi/2 - y$ therefore yields

$$\left(a - \frac{q}{a}\right) + \left(b - \frac{q}{b}\right) = \frac{(a+b)(ab-q)}{ab} \geq 0, \quad (11)$$

strictly for almost every y . Integrating proves $E(p) - qK(p) > 0$. Since $p+q > 1$, the bracket in (10) is strictly positive. Equations (7)–(10) show $H_\theta(1/2) > 0$, hence $H_\theta(\lambda) > 0$ for $0 \leq \lambda \leq 1/2$. Together with the negative- λ case and $s > 0$, this proves (3). $\square$

5 Full resolution of the conjecture

Theorem 2. *For every half-aperture $\omega \in (0, \pi)$, the MIT bag Dirac operator D_ω of Cassano–Lotoreichik is essentially self-adjoint. Its closure has exactly the H^1 domain in (1).*

Proof. For $0 < \omega < \pi/2$, this is Theorem 2.3 of the source [1]; for $\omega = \pi/2$, the half-space result recalled immediately after its Theorem 2.1 applies.

Let $\pi/2 < \omega < \pi$. The source proves that $\sigma(\mathbb{T}_k) \cap [-1/2, 1/2] = \emptyset$ for every $k \notin \{-1, 0\}$. It also proves that the two remaining modes are equivalent and that the $k = 0$ roots are precisely the solutions of (2). Lemma 1 shows that no such solution lies in $[-1/2, 1/2]$. Hence

$$\sigma(\mathbb{T}_k) \cap [-1/2, 1/2] = \emptyset \quad (k \in \mathbb{Z}).$$

The orthogonal fiber decomposition and one-dimensional limit-point criterion in Theorems 2.1 and 2.3 of the source now give essential self-adjointness and the domain formula (1). This is exactly the implication stated in Remark 2.4 of the source. $\square$

6 Verification, novelty, and limitations

The proof is analytic and does not rely on computation. The included high-precision verifier independently checks the Ferrers recurrence reduction, the Mehler–Dirichlet integral, the monotonic endpoint reduction, and the elliptic endpoint identity. It evaluates the source expression on 1,312 (ω, λ) pairs, including apertures within 10^{-6} of $\pi/2$ and near π , and returns PASS. See `VERIFIER.REPORT.md`.

A bounded novelty search on 12 August 2026 covered the four run indexes, arXiv, the published article page, exact-title and exact-conjecture searches, the author pair, and combinations of “MIT bag,” “non-convex cone,” “spin-orbit operator,” and “critical interval.” It found the source and its

published version, but no later paper resolving the all-aperture conjecture or proving the Ferrers inequality above. This is not a guarantee of priority.

The result resolves the circular-cone conjecture exactly as stated. It does not address general non-circular cones mentioned in Remark 2.5, nor does it extend the perturbative Hardy estimate of the source from convex to re-entrant cones. Human review should focus on the source-to-scalar reduction and the coefficient algebra leading from (6) to (7); the remaining endpoint argument is elementary.

References

- [1] B. Cassano and V. Lotoreichik, *Self-adjointness for the MIT bag model on an unbounded cone*, Math. Nachr. **297** (2024), 1006–1041; arXiv:2201.08192.
- [2] T. M. Dunster, *Legendre and Related Functions*, Chapter 14 in the NIST Digital Library of Mathematical Functions, especially Sections 14.9, 14.10, and 14.12, <https://dlmf.nist.gov/14>.